

JAVA STRING PROGRAMS

1. StringCharAt

Question / Aim

Write a Java program to find and display the character present at a specified index of a String using the charAt() method.

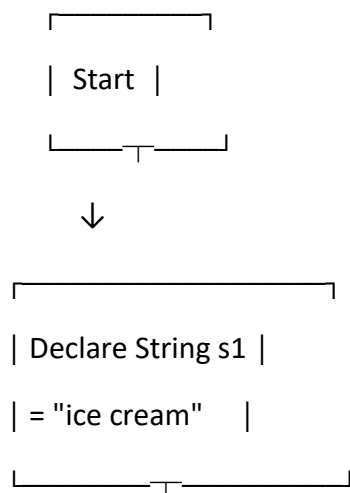
Steps for Execution

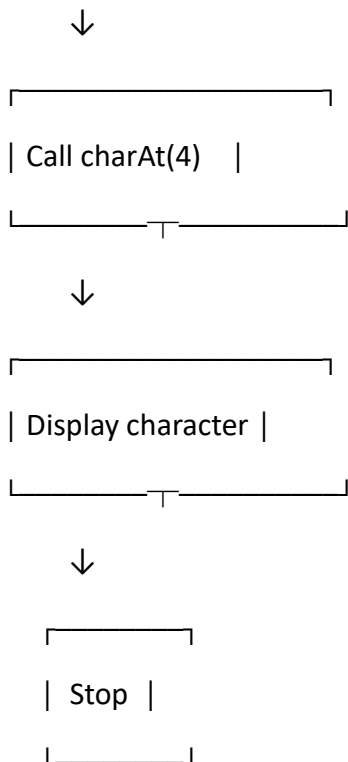
1. Open the Java IDE.
2. Create a package named javapractice.
3. Create a class named StringCharAt.
4. Declare the String "ice cream".
5. Use charAt(4) to obtain the character at index 4.
6. Display the character.
7. Compile and execute the program.

Algorithm

1. Start.
2. Declare a String s1 = "ice cream".
3. Call s1.charAt(4).
4. Store/display the returned character.
5. Stop.

UML Activity Diagram





Output

character=r

Exception Note

If the index is outside the String range, `StringIndexOutOfBoundsException` can occur.

2. StringConcat

Question / Aim

Write a Java program to concatenate two Strings using the `concat()` method.

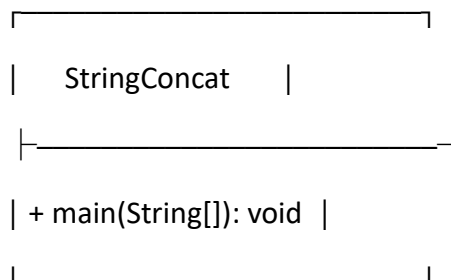
Steps for Execution

1. Create the class `StringConcat`.
2. Declare two Strings: "Hello " and "World".
3. Apply `concat()` to the first String.
4. Pass the second String as the argument.
5. Display the concatenated result.
6. Compile and execute.

Algorithm

1. Start.
2. Declare `s1 = "Hello "`.
3. Declare `s2 = "World"`.
4. Execute `s1.concat(s2)`.
5. Display the result.
6. Stop.

UML Class Diagram



Output

Result = Hello World

3. StringContains

Question / Aim

Write a Java program to check whether a String contains a specified sequence of characters using `contains()`.

Steps for Execution

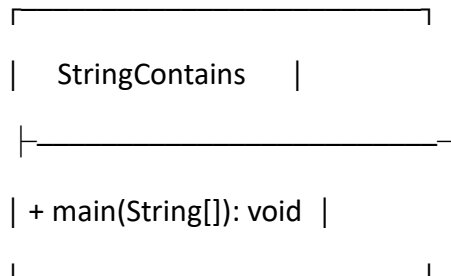
1. Create `StringContains`.
2. Declare the String `"Hello World"`.
3. Call `contains("World")`.
4. Display the Boolean result.
5. Compile and execute.

Algorithm

1. Start.
2. Declare `s = "Hello World"`.

3. Check whether s contains "World".
4. Display the result.
5. Stop.

UML Class Diagram



Output

Contains World = true

4. StringEndsWith

Question / Aim

Write a Java program to check whether a String ends with a specified suffix using endsWith().

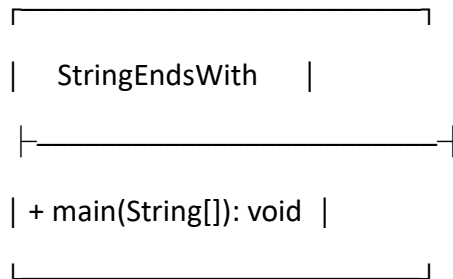
Steps for Execution

1. Create the class StringEndsWith.
2. Declare s = "java program".
3. Call endsWith("program").
4. Display the Boolean result.
5. Compile and execute.

Algorithm

1. Start.
2. Declare the String "java program".
3. Check whether the String ends with "program".
4. Display the result.
5. Stop.

UML Class Diagram



Output

Ends with program = true

5. StringEquals

Question / Aim

Write a Java program to compare two Strings using the `equals()` method.

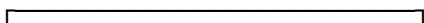
Steps for Execution

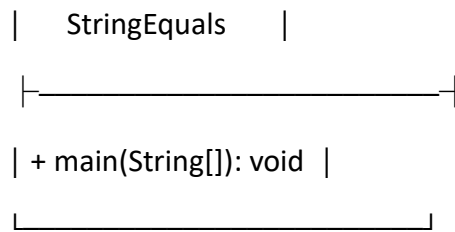
1. Create the class `StringEquals`.
2. Declare `s1 = "Hello"`.
3. Declare `s2 = "Hello"`.
4. Compare both Strings using `equals()`.
5. Display the result.
6. Compile and execute.

Algorithm

1. Start.
2. Initialize `s1` with `"Hello"`.
3. Initialize `s2` with `"Hello"`.
4. Execute `s1.equals(s2)`.
5. Display the result.
6. Stop.

UML Class Diagram





Output

Equal = true

6. StringEqualsIgnoreCase

Question / Aim

Write a Java program to compare two Strings without considering uppercase and lowercase differences using `equalsIgnoreCase()`.

Steps for Execution

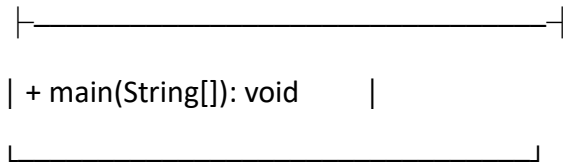
1. Create `StringEqualsIgnoreCase`.
2. Declare `s1 = "java program"`.
3. Declare `s2 = "JAVA PROGRAM"`.
4. Use `equalsIgnoreCase()`.
5. Display the result.
6. Compile and execute.

Algorithm

1. Start.
2. Declare the first String.
3. Declare the second String in uppercase.
4. Compare them using `equalsIgnoreCase()`.
5. Display the Boolean result.
6. Stop.

UML Class Diagram





Output

Equal = true

7. StringIndexOf

Question / Aim

Write a Java program to find the index position of a specified character in a String using `indexOf()`.

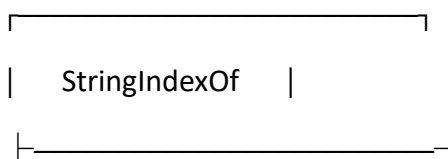
Steps for Execution

1. Create `StringIndexOf`.
2. Declare `s = "Hello"`.
3. Search for character 'l'.
4. Use `indexOf('l')`.
5. Display the returned index.
6. Compile and execute.

Algorithm

1. Start.
2. Declare `s = "Hello"`.
3. Search for 'l'.
4. Call `s.indexOf('l')`.
5. Display the index.
6. Stop.

UML Class Diagram



```
| + main(String[]): void |
```

```
|_____|
```

Output

Index = 2

Note: Java uses zero-based indexing, so the first l in "Hello" is at index 2.

8. StringIsEmpty

Question / Aim

Write a Java program to check whether a String is empty using the isEmpty() method.

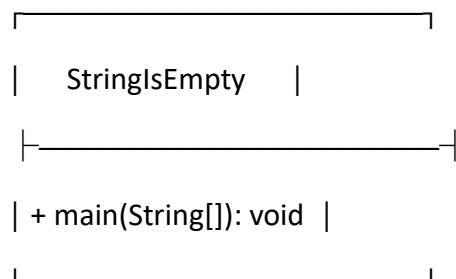
Steps for Execution

1. Create StringIsEmpty.
2. Declare s = "java".
3. Call isEmpty().
4. Display the result.
5. Compile and execute.

Algorithm

1. Start.
2. Declare the String "java".
3. Check whether the String has zero characters.
4. Display the result.
5. Stop.

UML Class Diagram



Output

Is Empty = false

9. StringLength

Question / Aim

Write a Java program to find the length of a String using the length() method.

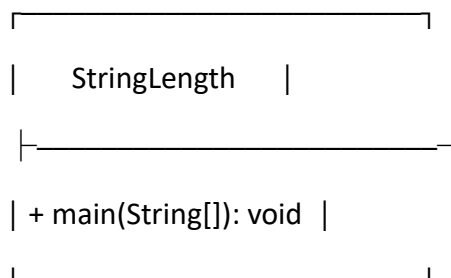
Steps for Execution

1. Create StringLength.
2. Declare s1 = "hello".
3. Call s1.length().
4. Display the returned length.
5. Compile and execute.

Algorithm

1. Start.
2. Declare s1 = "hello".
3. Calculate s1.length().
4. Display the length.
5. Stop.

UML Class Diagram



Output

length=5

10. StringLowerCase

Question / Aim

Write a Java program to convert a String into lowercase using toLowerCase().

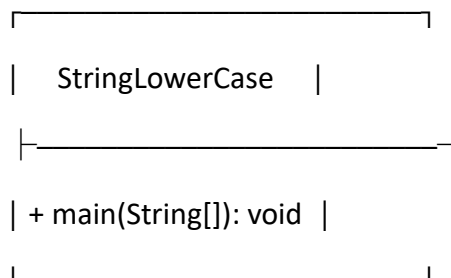
Steps for Execution

1. Create StringLowerCase.
2. Declare s = "HELLO".
3. Call toLowerCase().
4. Display the converted String.
5. Compile and execute.

Algorithm

1. Start.
2. Declare s = "HELLO".
3. Convert s to lowercase.
4. Display the result.
5. Stop.

UML Class Diagram



Output

Lowercase = hello

11. StringReplace

Question / Aim

Write a Java program to replace a character in a String using the replace() method.

Steps for Execution

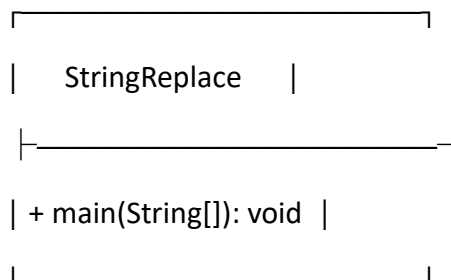
1. Create StringReplace.
2. Declare s = "javaprogram".

3. Replace character 'a' with 'x'.
4. Display the modified String.
5. Compile and execute.

Algorithm

1. Start.
2. Declare s = "javaprogram".
3. Call s.replace('a','x').
4. Display the returned String.
5. Stop.

UML Class Diagram



Output

Replaced = jxvxp rogrxm

More precisely, without the space:

Replaced = jxvxprogrxm

12. StringStartsWith

Question / Aim

Write a Java program to check whether a String starts with a specified prefix using startsWith().

Steps for Execution

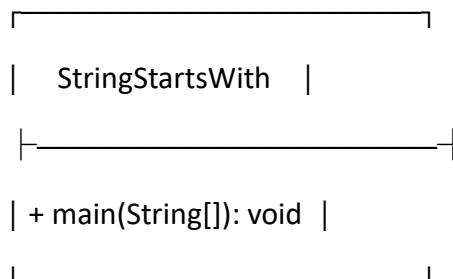
1. Create StringStartsWith.
2. Declare s = "ice cream".
3. Call startsWith("cream").

4. Display the result.
5. Compile and execute.

Algorithm

1. Start.
2. Declare s = "ice cream".
3. Check whether it starts with "cream".
4. Display the Boolean result.
5. Stop.

UML Class Diagram



Output

Starts with ice= false

Note: The actual code checks "cream", not "ice". Since "ice cream" starts with "ice", but not "cream", the output is false.

13. StringSubString

Question / Aim

Write a Java program to extract a portion of a String using the `substring()` method.

Steps for Execution

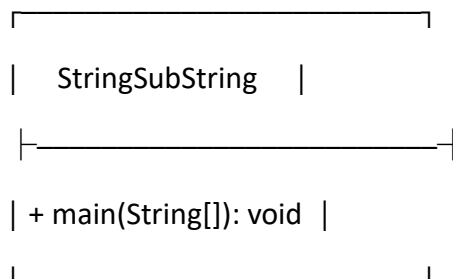
1. Create `StringSubString`.
2. Declare s = "Hello World".
3. Use `substring(6)`.
4. The substring starts from index 6.
5. Display the result.

6. Compile and execute.

Algorithm

1. Start.
2. Declare s = "Hello World".
3. Call s.substring(6).
4. Extract characters from index 6 to the end.
5. Display the substring.
6. Stop.

UML Class Diagram



Output

Substring = World

Exception Note

If the starting index is invalid, `StringIndexOutOfBoundsException` can occur.

14. StringTrim

Question / Aim

Write a Java program to remove leading and trailing whitespace from a String using the `trim()` method.

Steps for Execution

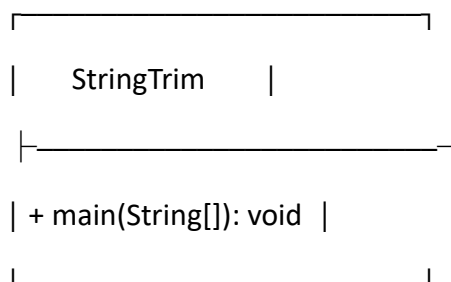
1. Create `StringTrim`.
2. Declare a String containing spaces before "Hello".
3. Call `trim()`.
4. Display the result.

5. Compile and execute.

Algorithm

1. Start.
2. Declare `s = " Hello"`.
3. Apply `trim()` to remove leading/trailing spaces.
4. Display the resulting String.
5. Stop.

UML Class Diagram



Output

After trim = Hello

15. StringUpperCase

Question / Aim

Write a Java program to convert a String into uppercase using the `toUpperCase()` method.

Steps for Execution

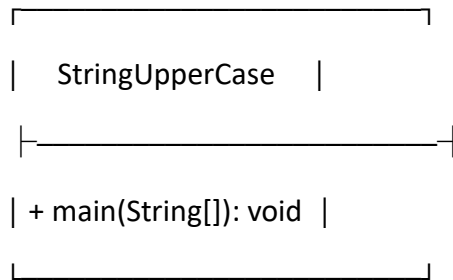
1. Create `StringUpperCase`.
2. Declare `s1 = "java program"`.
3. Call `toUpperCase()`.
4. Display the converted String.
5. Compile and execute.

Algorithm

1. Start.
2. Declare `s1 = "java program"`.

3. Convert it to uppercase using toUpperCase().
4. Display the result.
5. Stop.

UML Class Diagram



Output

uppercase=JAVA PROGRAM
